Multi-Agent System: Theories and Applications

Lecture 9: Formation Control of 1st and 2nd order Linear MASs

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Outline

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Formation Design

Assumption: $\mathcal{G}$ is undirected and connected

$$\dot{x}(t) = u(t)$$
$$x(t) \in \mathbb{R}^N, u(t) \in \mathbb{R}^N$$

Theorem 1

Formation is achieved with
$$u(t) = -\mu\mathcal{L}[x(t) - x^*]$$
for any $\mu > 0$, where $x^* \in \mathbb{R}^N$ such that $x_i^* - x_j^* = x_{ij}^*$ for a given desired formation pattern x_{ij}^*.

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How about with digraphs?

Application to Group of Front-Wheel Steering Vehicles

Assumption: $\mathcal{G}$ is undirected & connected

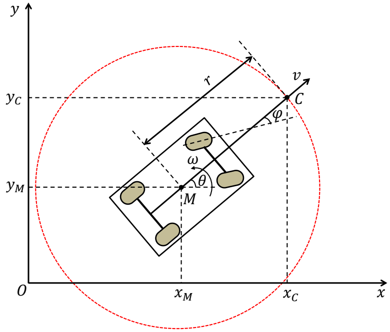
Mathematical model of vehicles:

$$\begin{aligned} \dot{\theta}_i &= \omega_i = v_i \tan \varphi_i \\ \dot{x}_{Mi} &= v_i \cos \theta_i \\ \dot{y}_{Mi} &= v_i \sin \theta_i \end{aligned}$$

Under a coordinate transform:

$$\dot{\bar{x}}_i = \bar{u}_i$$

with $\bar{x}_i \triangleq \begin{bmatrix} x_{Ci} \\ y_{Ci} \end{bmatrix}$, $M_i \triangleq \begin{bmatrix} \cos \theta_i & -r \sin \theta_i \\ \sin \theta_i & r \cos \theta_i \end{bmatrix}$,
and $\bar{u}_i \triangleq M_i \begin{bmatrix} v_i \\ \omega_i \end{bmatrix}$.



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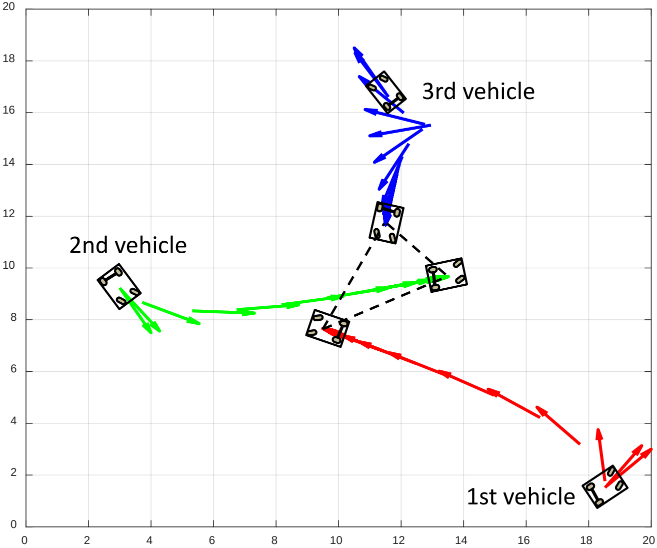
Simulation



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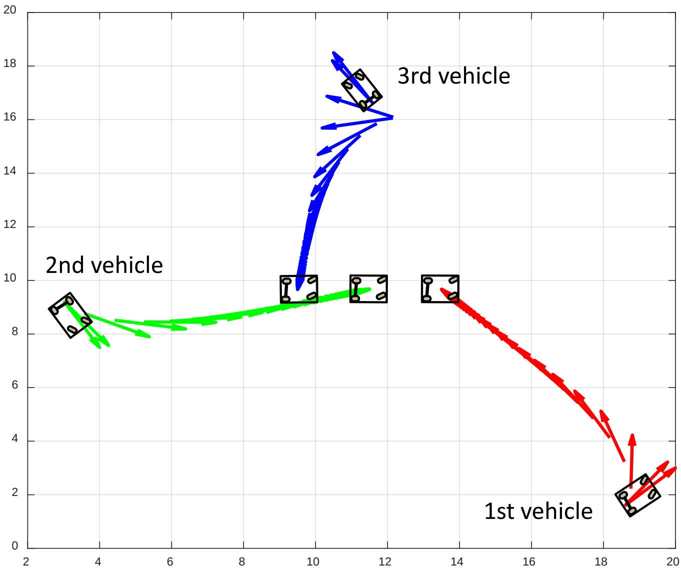
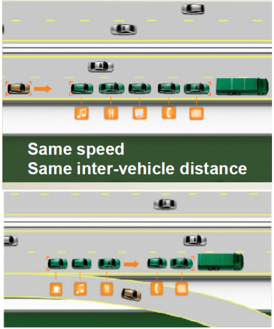

3rd vehicle
2nd vehicle
1st vehicle
20
18
16
14
12
10
8
6
4
2
0
2
4
6
8
10
12
14
16
18
20

Formation Control Law



Assumption: $\mathcal{G}$ is undirected and connected

$$\dot{x}(t) = (I_N \otimes A)x(t) + (I_N \otimes B)u(t)$$

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Theorem 2

Formation is achieved with

$$u(t) = -\left(\mathcal{L} \otimes \begin{bmatrix} k_1 & k_2 \end{bmatrix}\right)(x(t) - x^*)$$

where $x^* \in \mathbb{R}^{2N}$ such that $x_i^* - x_j^* = x_{ij}^*$ for a given desired formation pattern x_{ij}^*, and $K \triangleq \begin{bmatrix} k_1 & k_2 \end{bmatrix}$ such that $A - \lambda_i BK$ is stable $\forall\, i = 2, \ldots, N$, in which $\lambda_i, i = 2, \ldots, N$ are non-zero eigenvalues of $\mathcal{L}$.

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Formation Control Law

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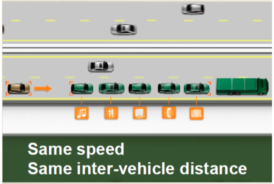

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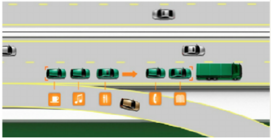

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How about with digraphs?

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Other Formation Control Designs

- Generic dynamics of agents
- Distance-based formation control
- Rigid formation control
- Limited-information formation control, e.g., position-based only, angle-based only, etc.